# TOPICS COVERED

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# Case Study in Brief

This project focuses on the development of an intelligent meal-planning solution designed specifically for time-constrained users. The core objective is to eliminate the daily friction of "what to eat" by streamlining the transition from a state of indecision to a structured weekly menu. By leveraging three primary data pillars—**time availability, dietary requirements, and existing kitchen inventory**—the tool transforms meal planning from a chore into a seamless, data-driven experience. Developed using the Lovable framework and refined through iterative user testing, this prototype addresses the logistical and cognitive load that often prevents busy individuals from maintaining consistent, healthy, and cost-effective eating habits.

# User Research Summary (with proof)

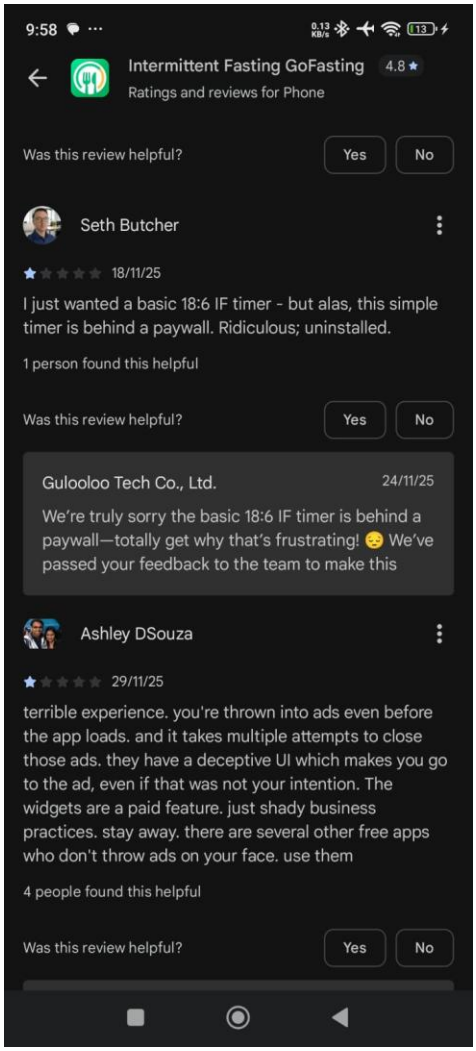
This section outlines why current meal-planning apps are failing and what users actually need to stay consistent. While many apps exist, our research shows that most fail to use AI in a way that truly helps the user, leaving a "gap" in the market that our tool aims to fill.

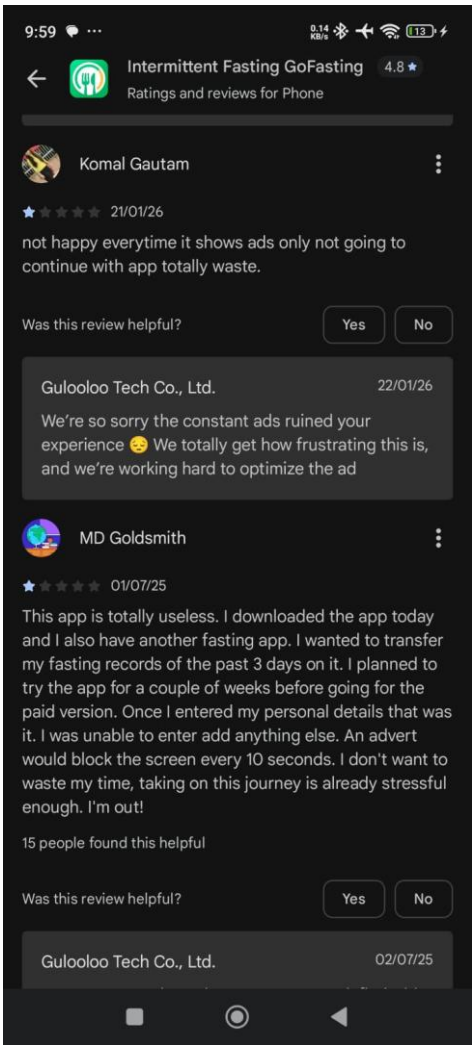
## Core Research Findings:

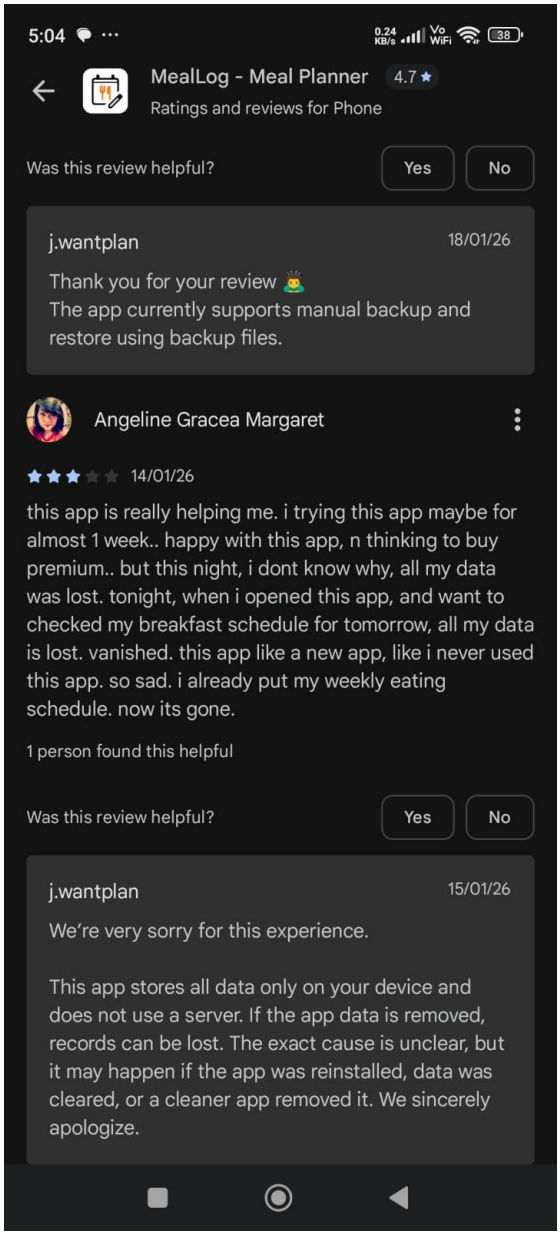
To make a meal planner that users actually want to keep using, our research identified five non-negotiable requirements:

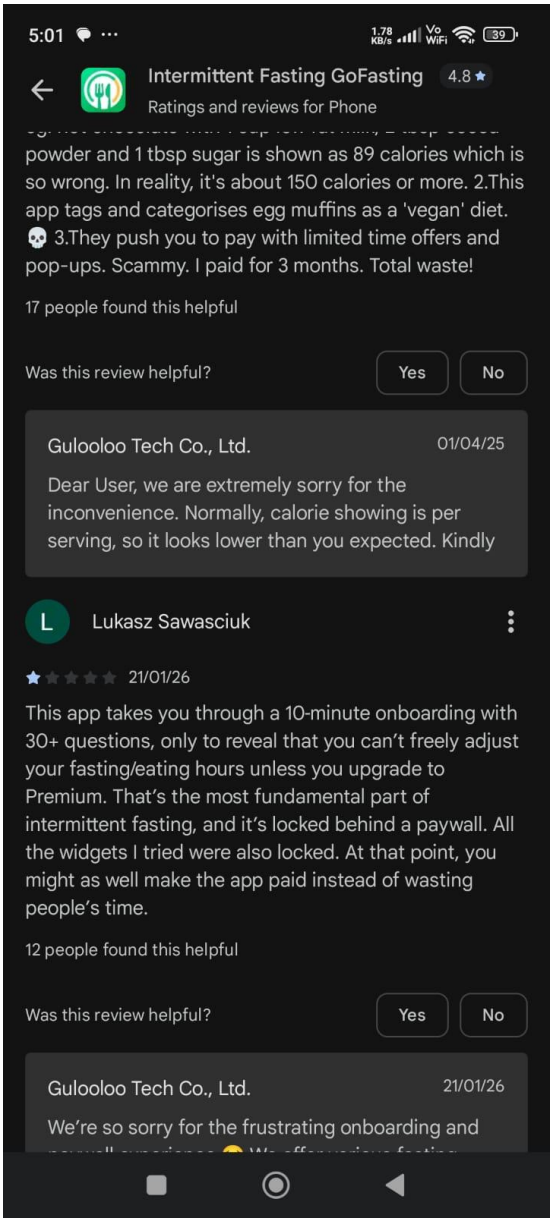
- **Ad-Light Experience:** Users are easily frustrated by constant interruptions. To keep them focused on planning, the app must strictly limit ad redirections.
- **Speedy Onboarding:** First impressions matter. Users want to start planning immediately rather than spending 10 minutes filling out a long profile.
- **Seamless Interface:** A "glitch-free" and simple UI is essential. If the app feels clunky or confusing, users will revert to using simple pen and paper.

- **Reliable Data Syncing:** By using Backend Cloud Storage, we ensure that a user’s recipes, inventory, and schedules are always saved and accessible on any device.
- **Adaptive Scheduling:** Meal plans shouldn't be rigid. Users need a flexible system where they can easily move or change meals when their daily schedule shifts unexpectedly.






## The 'Satisfactory Gap'(Conclusion):

Our research proved that while many apps have "AI," they aren't using it to solve the user's specific problems. Users don't want a complex tool; they want a smart assistant that handles the heavy lifting—like tracking ingredients and managing time—so they don't have to.

# Pain Points and Problem Statements

This section breaks down the specific technical and experiential failures of existing solutions that prevent users from reaching their meal-planning goals.

## Core User Pain Points:

- **Vanishing Data:** Users lose weeks of work because many apps save data only on the phone. If the app glitches or is deleted, all their meal plans are gone.
- **Trust Issues with Math:** Calorie counts are often wrong. This makes users confused and discouraged when they don't see the health results they expected.
- **The "Pay-to-Play" Wall:** Basic features that should be free (like making a simple healthy schedule) are locked. This feels like a "bait and switch" to new users.
- **Ad Exhaustion:** Every click triggers a long ad. This makes a 2-minute task (checking tonight's dinner) take 10 minutes.
- **"Trap" Buttons:** Deceptive designs trick users into clicking ads that send them to the Play Store or random websites, making the experience feel "terrible" and unsafe.
- **Starting Over After Paying:** Even if a user buys the premium version, they are often forced to redo their setup or learn a brand-new layout, which is exhausting.

## The Problem Statements:

**1. The "Permanent Progress" Problem:** Busy users need a way to save their meal plans to a secure "cloud" account so that their data is never lost and is always ready whenever they switch devices.

**2. The "Accuracy & Integrity" Problem:** Health-conscious users need a tool they can trust, providing accurate nutritional math that updates automatically so they don't have to second-guess their health goals.

**3. The "Respect for Time" Problem:** Everyday users need a clean, fast experience that lets them organize their kitchen and schedule meals without being tricked by deceptive ads or forced into a complicated setup process.

# Key User Flows

This section visualizes the specific navigation paths within the NutriGen interface. The flows are designed to minimize "time-to-value," ensuring users can move from opening the app to having a personalized plan with as few taps as possible.

## Flow 1: Intelligent Onboarding and Goal Setting

*Goal: Establish a secure account and personal health baseline.*

- **Quick Start:** The user lands on a clean welcome screen and chooses a one-tap sign-in method (Google/Email).
- **Goal Identification:** A simple selection screen appears where the user chooses their primary objective (e.g., "Weight Loss," "Build Muscle," or "Maintain Balance").
- **Dietary Guardrails:** The user selects any restrictions (e.g., "Vegetarian," "Gluten-Free") through a "chip-selection" interface.
- **Automatic Backup:** Upon completion, the app displays a "Sync Successful" indicator, confirming that the profile is now stored in Backend Cloud Storage.

## Flow 2: Pantry-to-Plate AI Generation

*Goal: Generate a recipe based on available time and ingredients.*

- **Inventory Log:** From the main dashboard, the user enters the "My Pantry" section and quickly checks off ingredients they already have.
- **Time Input:** A sliding scale or quick-select buttons allow the user to set their cooking window (e.g., "15 Minutes" or "30 Minutes").

- **The AI Prompt:** The user taps "Generate Smart Meal." The NutriGen AI processes the inventory and time constraints to suggest a specific dish.
- **Accept & Schedule:** The user views the recipe and taps "Add to Weekly Plan," which instantly populates their calendar.

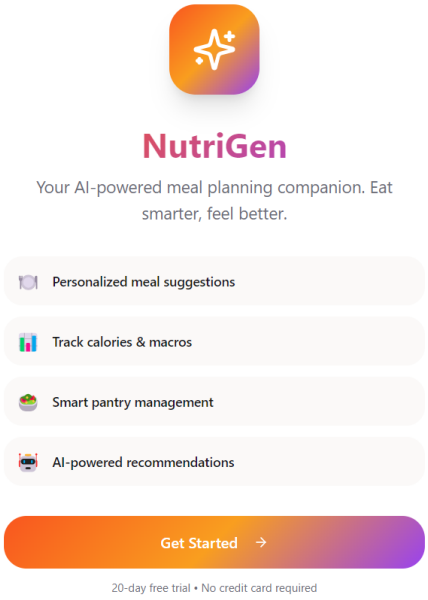
## Flow 3: Accurate Tracking and Schedule Management

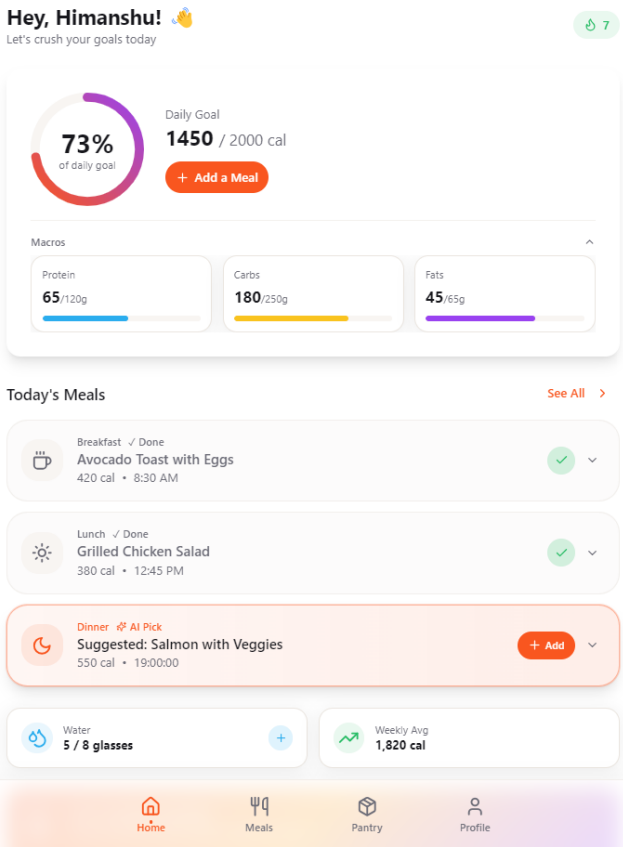
*Goal: View nutritional facts and adjust the week's schedule.*

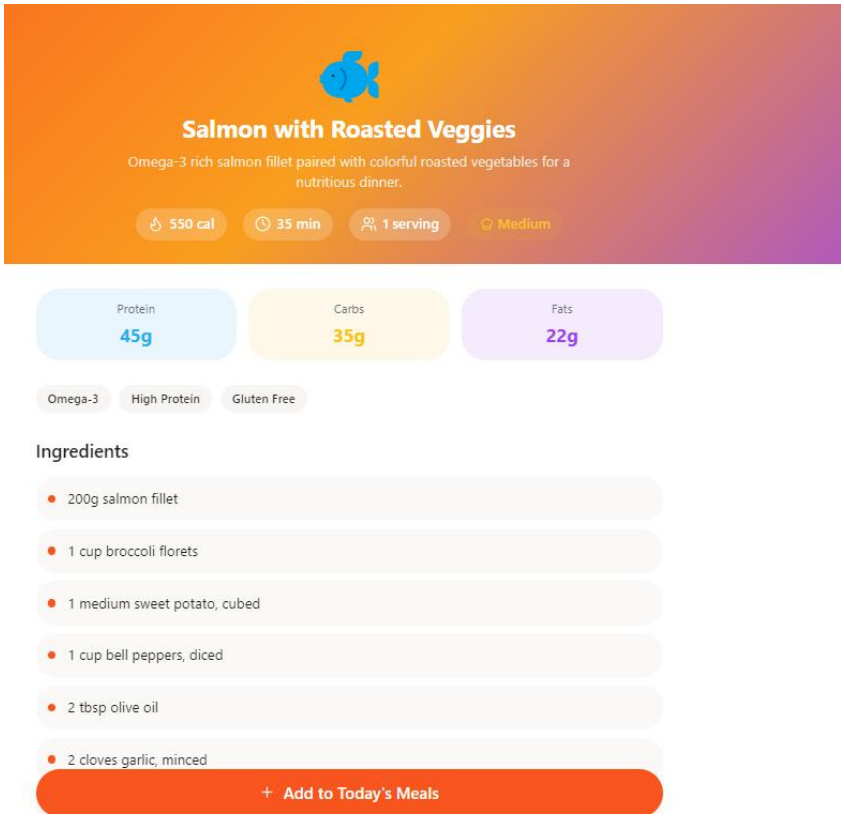
- **Dashboard Overview:** The user opens the "Weekly Planner" view, which shows a day-by-day breakdown of scheduled meals.
- **Nutritional Deep-Dive:** Tapping a meal card opens a detailed view showing a verified calorie count and macro breakdown (Protein/Carbs/Fats), addressing the "accuracy" pain point.
- **Flexible Rescheduling:** To change plans, the user uses a long-press gesture to drag a meal from one day to another.
- **Real-Time Recalculation:** The app instantly updates the total daily calories for both days to reflect the move, ensuring the user stays within their target range.

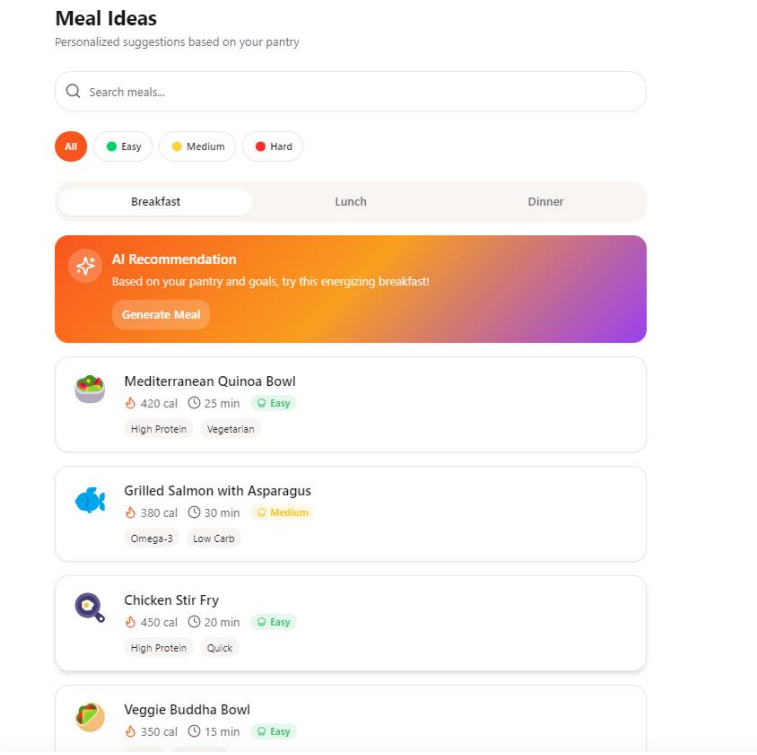
# NutriGen Prototype UI

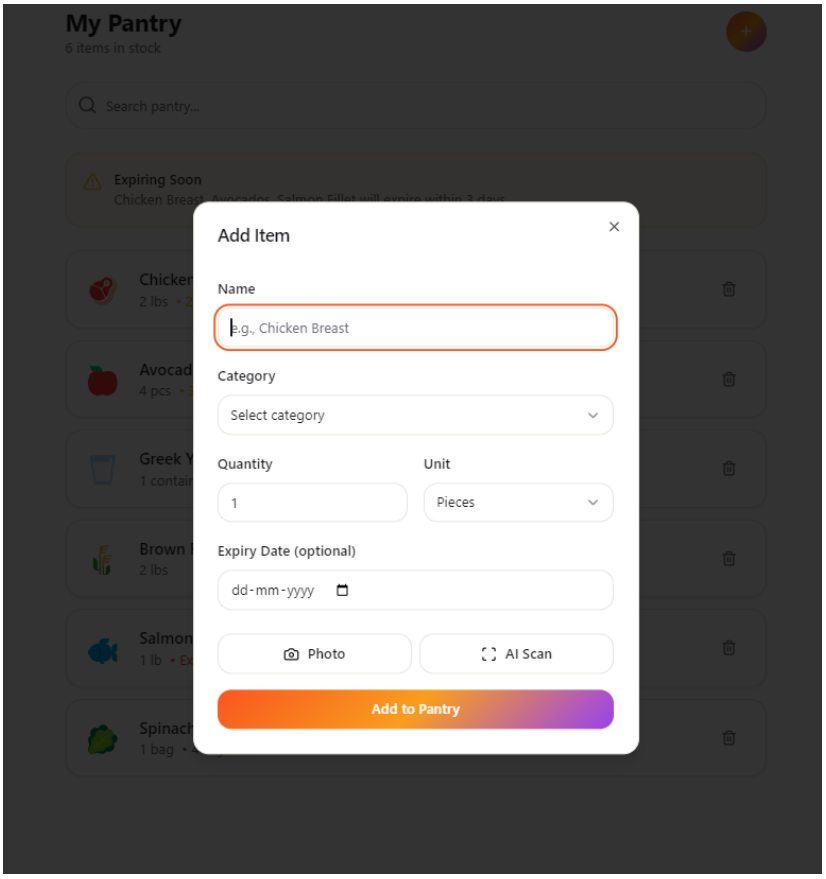
## Screenshots:

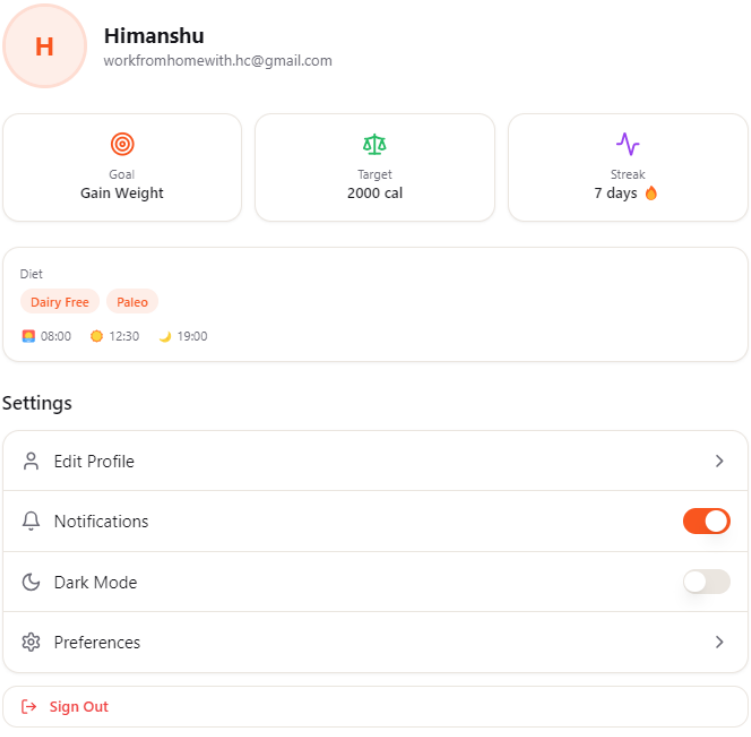



Salmon with Roasted Veggies
Omega-3 rich salmon fillet paired with colorful roasted vegetables for a nutritious dinner.
550 cal
35 min
1 serving
Medium
Protein
45g
Carbs
35g
Fats
22g
Omega-3
High Protein
Gluten Free
Ingredients
200g salmon fillet
1 cup broccoli florets
1 medium sweet potato, cubed
1 cup bell peppers, diced
2 tbsp olive oil
2 cloves garlic, minced
Add to Today's Meals


Meal Ideas
Personalized suggestions based on your pantry
Search meals...
All
Easy
Medium
Hard
Breakfast
Lunch
Dinner
AI Recommendation
Based on your pantry and goals, try this energizing breakfast!
Generate Meal
Mediterranean Quinoa Bowl
420 cal
25 min
Easy
High Protein
Vegetarian
Grilled Salmon with Asparagus
380 cal
30 min
Medium
Omega-3
Low Carb
Chicken Stir Fry
450 cal
20 min
Easy
High Protein
Quick
Veggie Buddha Bowl
350 cal
15 min
Easy


My Pantry
6 items in stock
Search pantry...
Expiring Soon
Add Item
Name
e.g., Chicken Breast
Category
Select category
Quantity
1
Unit
Pieces
Expiry Date (optional)
dd-mm-yyyy
Photo
AI Scan
Add to Pantry


H
Himanshu
workfromhomewith.hc@gmail.com
Goal
Gain Weight
Target
2000 cal
Streak
7 days
Diet
Dairy Free
Paleo
08:00
12:30
19:00
Settings
Edit Profile
Notifications
Dark Mode
Preferences
Sign Out

These are some of the UI Screens for the NutriGen App prototype. Although, most of the reviews were very positive but some feedbacks were on onboarding and simplified view in the home screen so I made some additions in onboarding and made it flexible + I made the home screen look cleaner and better. I Hope you will like that too.

## Preview Link:

https://nutri-gen-app.lovable.app

The webapp prototype is Still undergoing development and updates.

## Core Features:

- Smart Onboarding & Auto Calorie Targeting Multi-step profile setup (metrics, activity, goals, diet, meal times) that auto-calculates a personalized daily calorie target.
- AI Meal Suggestions Personalized recipe recommendations based on user goals, dietary preferences, and allergies and convenience
- Pantry-Aware AI Recipes Generates meal ideas from ingredients the user already has in their pantry, reducing food waste.
- Budget Mode Weekly budget input that tailors meal suggestions to cost-conscious users.
- Meal Logging & History Log meals by type with full macro breakdown and view a chronological history.
- Pantry Management with Expiry Tracking Inventory ingredients with quantity, category, and expiry dates to minimize waste.
- Live-Sync Preferences Edits to goals, diet, or meal times in Preferences instantly reflect on the Profile and Dashboard.
- Secure Profile Management Edit username, avatar, email & mobile with OTP verification, plus trial/subscription status tracking.

# How the NutriGen Prototype Solves Core Problems

| Identified Problem | The NutriGen Solution | The User Benefit |
|---|---|---|
| **Data Loss (Storage)** | **Integrated Cloud Syncing:** Every action (adding a recipe or marking a meal) is instantly saved to the cloud rather than local storage. | Users never lose progress, even if they switch devices or delete the app. |
| **Incorrect Calorie Counts** | **Verified Recipe Database:** The AI only draws from a verified library of nutritional data to ensure "math integrity." | Users can trust the app for their health goals without needing to double-check. |
| **Ad and UI Frustration** | **Clean, Purposeful UI:** Deceptive "trap" buttons and full-screen ad redirections have been removed in favour of a smooth, professional interface. | Tasks take seconds instead of minutes; the "respect for time" creates a premium feel. |
| **Rigid Scheduling** | **Drag-and-Drop Interaction:** The calendar is fully interactive, allowing for instant changes when life gets busy. | Users don't feel "guilty" or "stuck" when a plan changes; they simply move the meal. |
| **Paywall Frustration** | **Value-First Access:** Basic healthy scheduling and calorie tracking are unlocked from Day 1. | Users experience the app's value before being asked to subscribe for advanced features. |




## Made By:

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